

Question ID: 2f958af9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$v^2 = \frac{LT}{m}$$

The formula above expresses the square of the speed v of a wave moving along a string in terms of tension T , mass m , and length L of the string. What is T in terms of m , v , and L ?

Answer

- A. $T = \frac{mv^2}{L}$
- B. $T = \frac{m}{v^2L}$
- C. $T = \frac{mL}{v^2}$
- D. $T = \frac{L}{mv^2}$

Correct Answer: A

Rationale

Choice A is correct. To write the formula as T in terms of m , v , and L means to isolate T on one side of the equation. First, multiply both sides of the equation by m , which gives $mv^2 = \frac{mLT}{m}$, which simplifies to $mv^2 = LT$. Next, divide both sides of the equation by L , which gives

$$\frac{mv^2}{L} = \frac{LT}{L}, \text{ which simplifies to } T = \frac{mv^2}{L}.$$

Choices B, C, and D are incorrect and may be the result of incorrectly applying operations to each side of the equation.